



COS20028 – BIG DATA ARCHITECTURE AND APPLICATION

Dr Pei-Wei Tsai (Unit Convenor)
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Welcome to COS20028

Big Data Architecture and Application

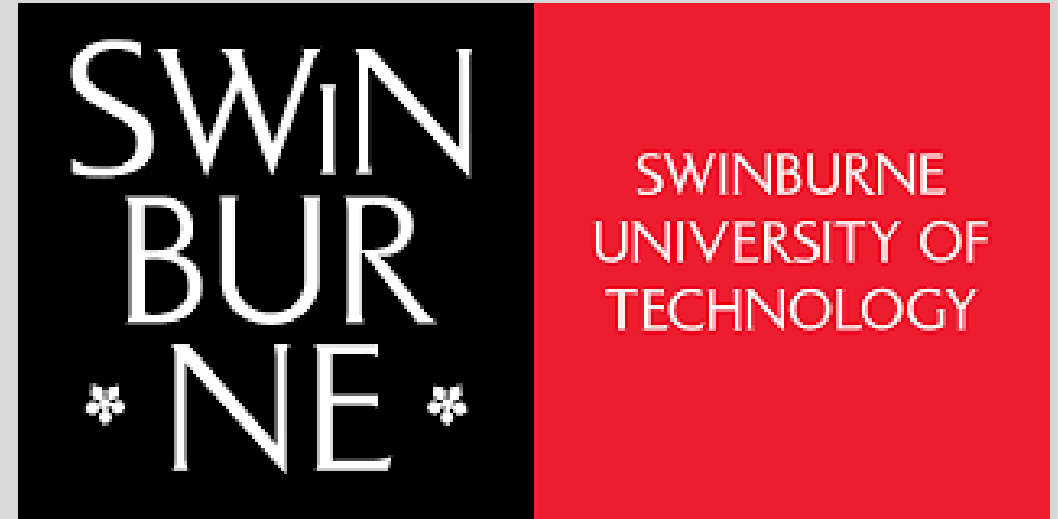
Asynchronised
Short Videos

Live Online
Lectures

Lab Classes

Online Tests
(week 4, 7, and 12)

Assignments
(Week 5 and 11)



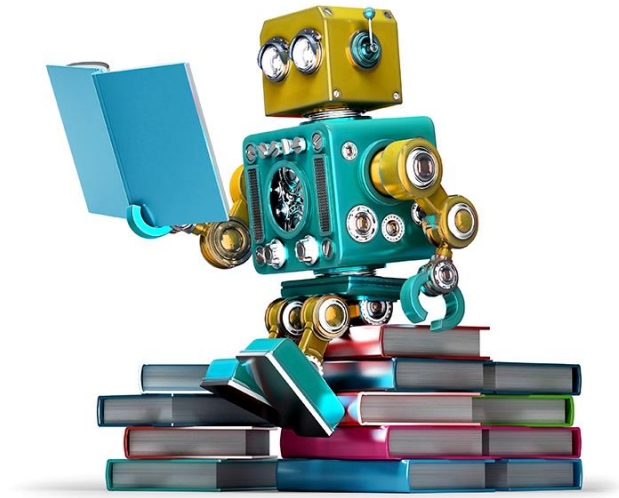
- **Lecturer and Convenor**
 - **Dr Pei-Wei Tsai**
ptsai@swin.edu.au
Office: EN508d
 - **Main Research Fields:**
 - **Intelligent Optimisation**
 - **Data Analytics**
 - **Machine Learning**
 - **Differential Privacy**



ABOUT THIS UNIT (COS20028)

Learning Outcomes

- After successfully completing this unit, you should be able to:
1. Appreciate the architecture of Hadoop clusters at both hardware and system software levels.
 2. Demonstrate the skills in working with the open source Hadoop platform through the enterprise solutions.
 3. Apply Hadoop and other related Big data technologies such as MapReduce, Hive, Impala, Pig in developing analytics and solving common problems faced by enterprises today.





Lab Environment



Hardware

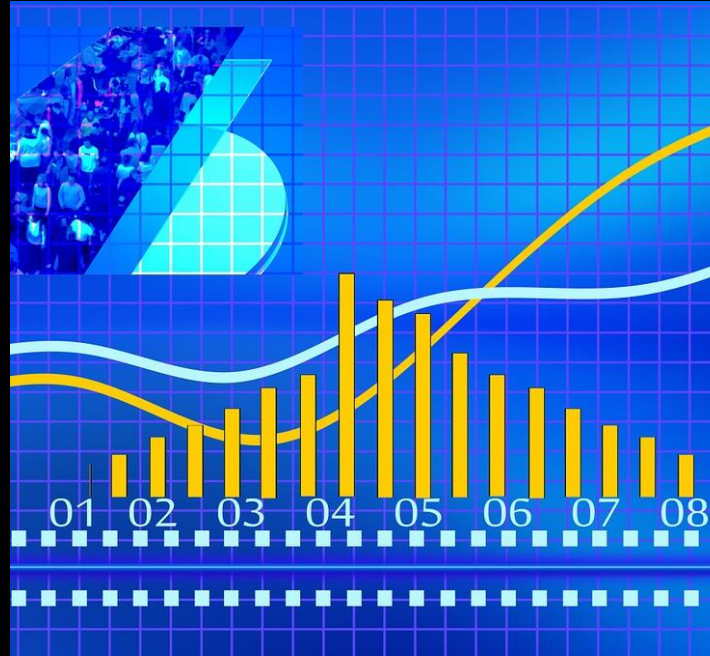
A PC or a laptop.



Software

VMware
Workstation
Player or Oracle
VM VirtualBox
Secure FTP
Product (Optional)

A hand holding a red marker, writing the word "analysis" in red cursive script over a blurred background of a bar chart.



```

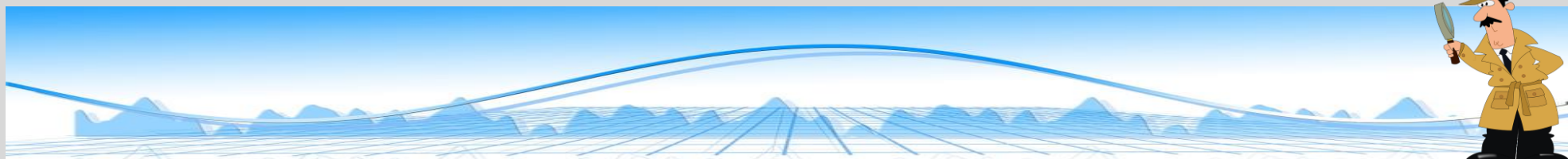
78 $SESSION['CAPTCHA']['config'] = serialize($captcha_config);
79
80 return array(
81     'code' => $captcha_config['code'],
82     'image_src' => $image_src
83 );
84
85 }
86
87
88 if (function_exists('hex2rgb')) {
89     function hex2rgb($hex_str, $return_string = false, $separator = ',') {
90         $hex_str = preg_replace("/[A0-9A-Fa-f]/", '', $hex_str); // Ge
91         $rgb_array = array();
92         if (strlen($hex_str) == 6) {
93             $color_val = hexdec($hex_str);
94             $rgb_array['r'] = 0xFF & ($color_val >> 0x10);
95             $rgb_array['g'] = 0xFF & ($color_val >> 0x08);
96             $rgb_array['b'] = 0xFF & ($color_val >> 0x00);
97         } elseif (strlen($hex_str) == 3) {
98             $rgb_array['r'] = hexdec(str_repeat(substr($hex_str, 0, 1), 2));
99             $rgb_array['g'] = hexdec(str_repeat(substr($hex_str, 1, 1), 2));
100             $rgb_array['b'] = hexdec(str_repeat(substr($hex_str, 2, 1), 2));
101         } else {
102             return false;
103         }
104         return $return_string ? implode($separator, $rgb
105     }
106 }
107
108 // Draw the image
109 if (isset($_GET['code'])) {
110     $code = $_GET['code'];
111     $image_src = $SESSION['CAPTCHA']['image_src'];
112     $image_data = file_get_contents($image_src);
113     $image = imagecreatefromstring($image_data);
114     $text_color = $SESSION['CAPTCHA']['text_color'];
115     $font_size = $SESSION['CAPTCHA']['font_size'];
116     $font = $SESSION['CAPTCHA']['font'];
117     $text = $code;
118     $text_x = 5;
119     $text_y = 5;
120     $text_w = strlen($text) * $font_size;
121     $text_h = $font_size;
122     $text_x2 = $text_x + $text_w;
123     $text_y2 = $text_y + $text_h;
124     $text_x_c = ($text_x2 - $text_x) / 2;
125     $text_y_c = ($text_y2 - $text_y) / 2;
126     $text_w_c = ($text_x2 - $text_x) / 2;
127     $text_h_c = ($text_y2 - $text_y) / 2;
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Expected Background

- **Objective-oriented Programming Ability.**
- **Basic Java Programming.**
- **Regular Expression.**
- **Background knowledge in Data Science.**

Assessment and Task Details

Assessment Task	Task Type	Weighting	Submission Due Date
Assignment 1	Individual	20%	Week 05
Assignment 2		20%	Week 11
Test 1		15%	Week 04 Class
Test 2		15%	Week 07 Class
Test 3		30%	Week 12 Class





**DEADLINE
* EXTENDED ***

Special Consideration?

If you encounter some difficulties such as medical issues, which affects your progress, during the study, you'll need to launch a special consideration.

The special consideration will only extend the deadline in a reasonable range for submitting the assignments. It will not be used to twist the evaluation criteria.

A valid medical certificate should be issued by any authorised Australian GP and clearly state the patient, and the impact period of unfit for study.



**DEADLINE
*EXTENDED***

Special Consideration?

A special exam is offered to students who are granted special considerations, unless it is possible to accommodate the special considerations during the semester.



Before the end of the semester: deadline extended.
After the end of the semester: a special exam is required.

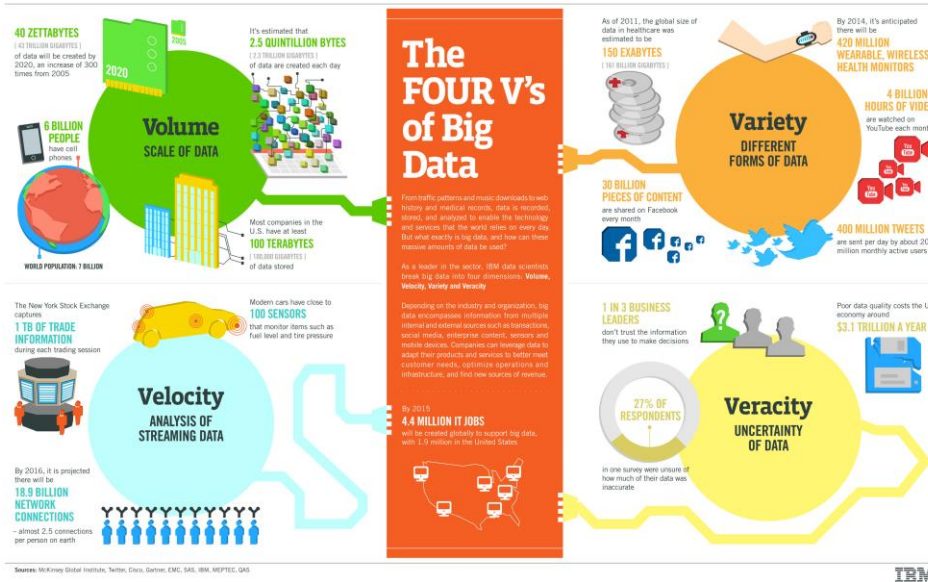
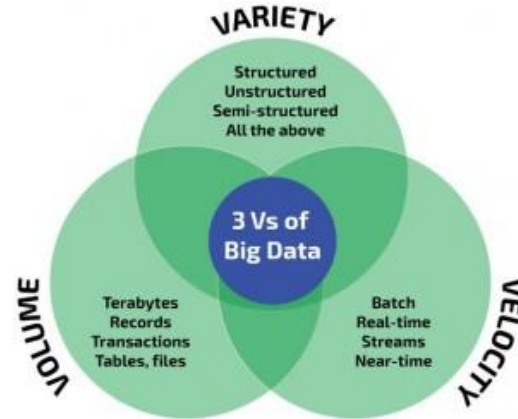


THE PLATFORM FOR STORING BIG DATA

Big Data

- Exceeds the processing capacity of conventional database systems
 - Too big (TB/PB level)
 - Moves too fast
 - New structure
- Four Vs of Big Data:
 - Volume
 - Velocity
 - Variety
 - (Veracity)

Quantities of bytes						
Common prefix				Binary prefix		
Name	Symbol	Decimal	Binary	Name	Symbol	Binary
		SI	JEDEC			IEC
kilobyte	KB/kB	10 ³	2 ¹⁰	kibibyte	KiB	2 ¹⁰
megabyte	MB	10 ⁶	2 ²⁰	mebibyte	MiB	2 ²⁰
gigabyte	GB	10 ⁹	2 ³⁰	gibibyte	GiB	2 ³⁰
terabyte	TB	10 ¹²	2 ⁴⁰	tebibyte	TiB	2 ⁴⁰
petabyte	PB	10 ¹⁵	2 ⁵⁰	pebibyte	PiB	2 ⁵⁰
exabyte	EB	10 ¹⁸	2 ⁶⁰	exbibyte	EiB	2 ⁶⁰
zettabyte	ZB	10 ²¹	2 ⁷⁰	zebibyte	ZiB	2 ⁷⁰
yottabyte	YB	10 ²⁴	2 ⁸⁰	yobibyte	YiB	2 ⁸⁰



IBM



Moving from the regular data to the big data realm?

- How much data an average person generates per day?
 - Mobile phone geo-location data.
 - Social media data.
 - Communications.
 - IoT devices.
 - ...
- With proper data processing techniques, new knowledge or information can be extracted from the data generated by people in their daily life.



BIG DATA ECOSYSTEM

Big Data



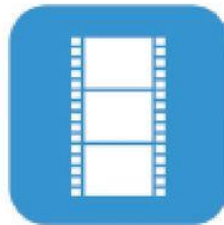
Mobile
Sensors



Social
Media



Video
Surveillance



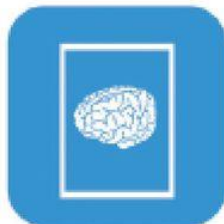
Video
Rendering



Smart
Grids



Geophysical
Exploration



Medical
Imaging

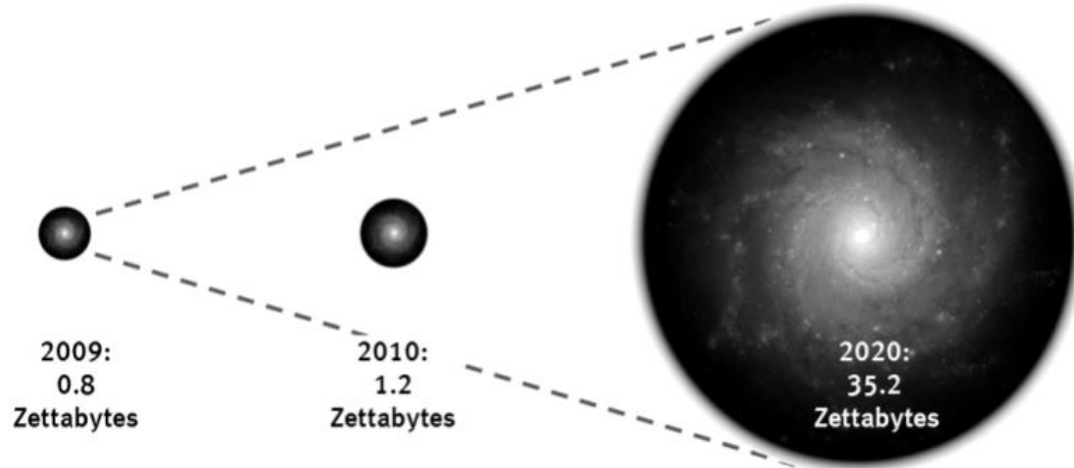


Gene
Sequencing

DRIVERS OF BIG DATA

Big Data Size: The Volume Of Data Continues To Explode

The Digital Universe 2009 - 2020

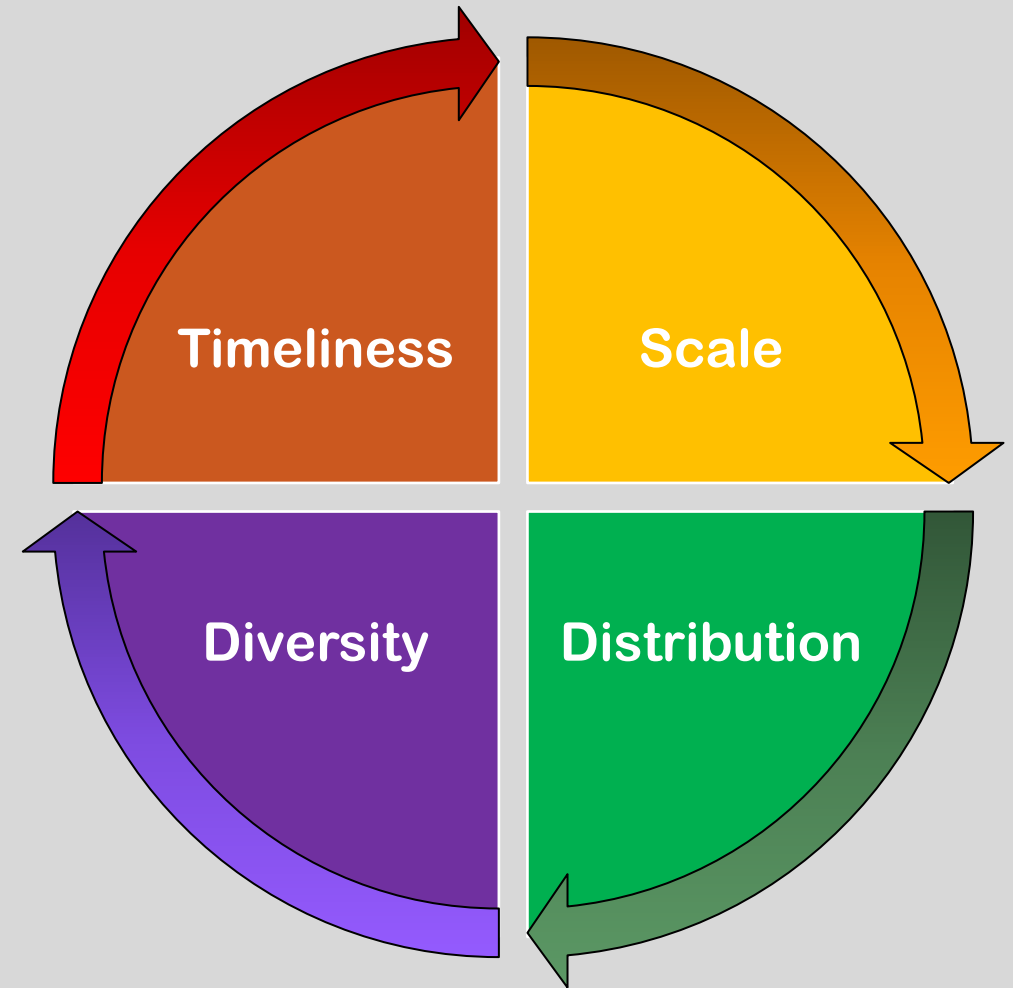


Key Characteristics of Big Data

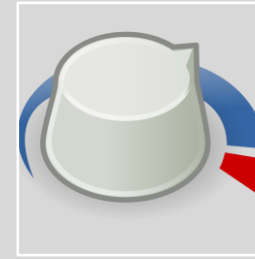
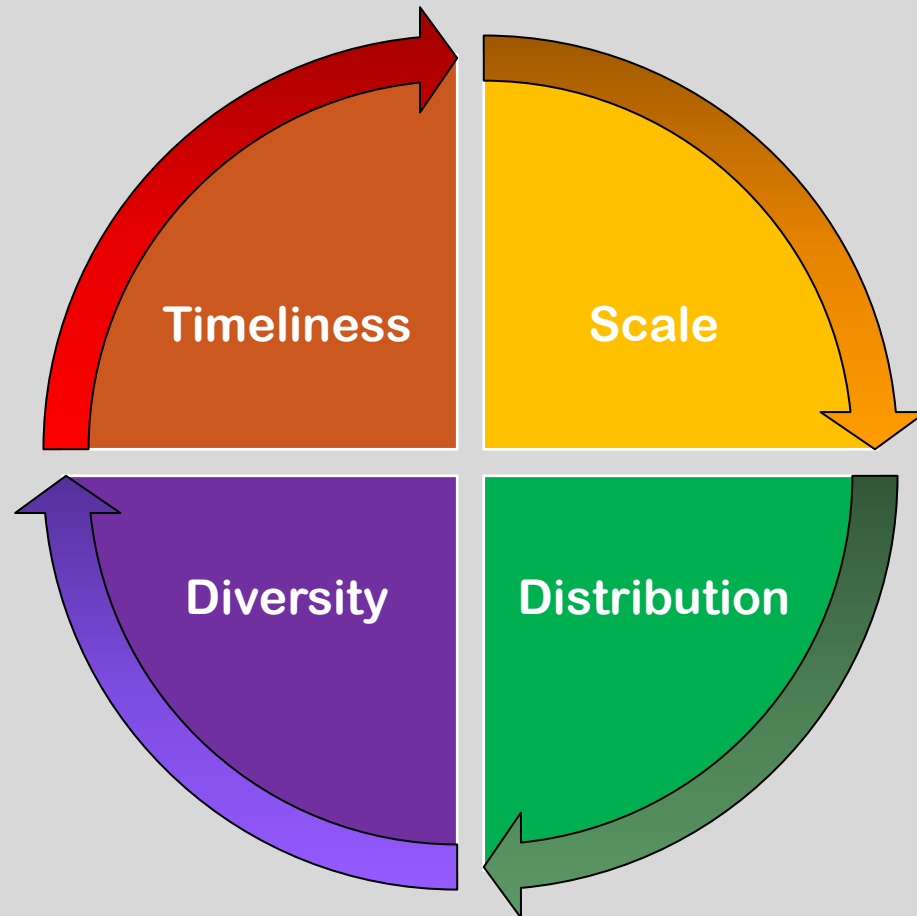
- **Data Volume**
 - 44x increase from 2010 to 2020. (1.2 Zetta Bytes to 35.2 ZB)
- **Processing Complexity**
 - Changing data structures.
 - Use cases warranting additional transformations and analytical techniques.
- **Data Structure**
 - Greater variety of data structures to mine and analyse.

Big Data: What is

- Big Data is data whose scale, distribution, diversity, and/or timeliness require the use of new technical architectures and analytics to enable insights that unlock new sources of business value.
 - by M. James, C. Michael, B. Brad, B. Jacques, D. Richard, R. Charles, and H. Angela, 2011. Big data: the next frontier for innovation, competition, and productivity. *The McKinsey Global Institute*.



Big Data 4 Vs Revisit

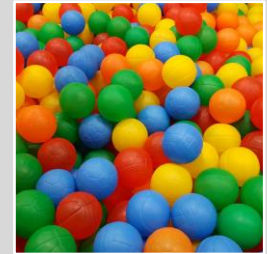


Volume

- Scale

Variety

- Diversity
- Distribution

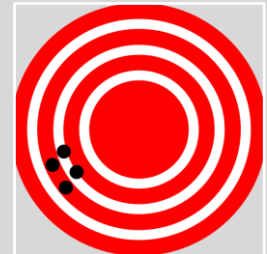


Velocity

- Timeliness

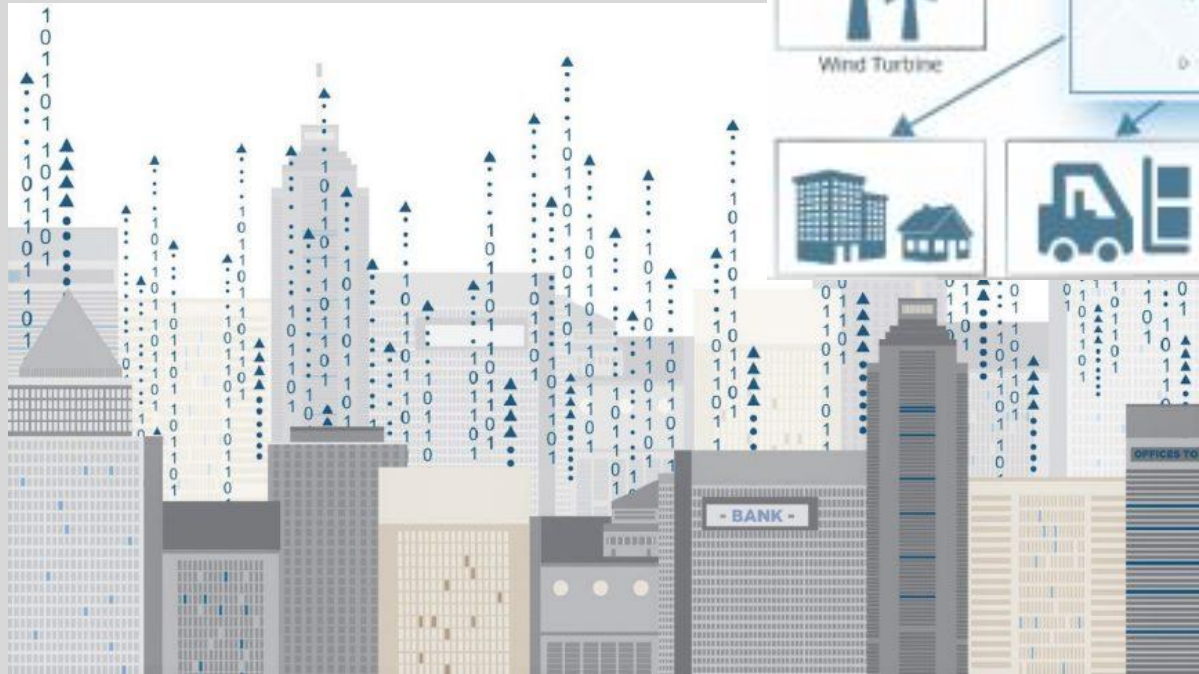
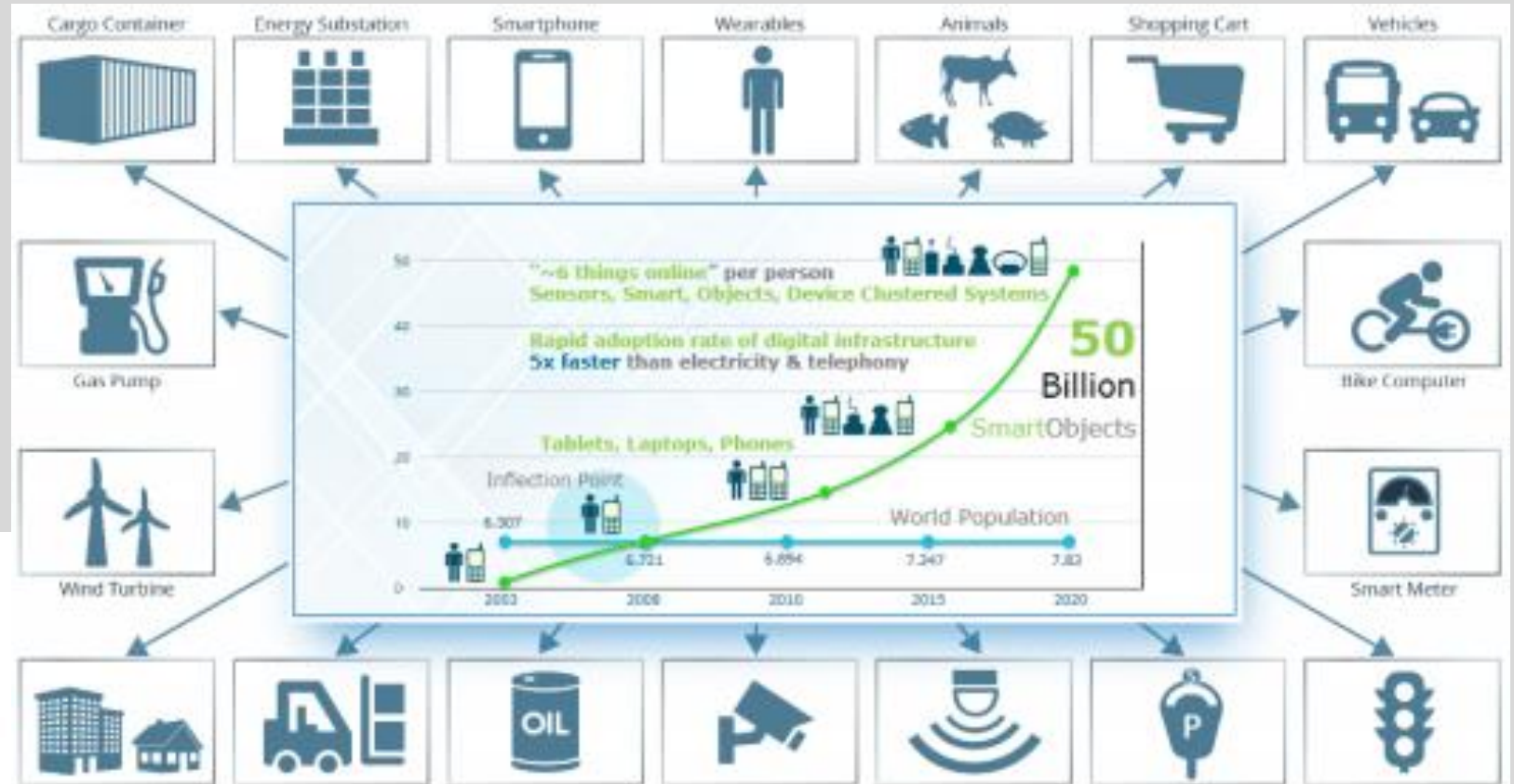
Veracity

- i.e. pertaining to the accuracy of data.



Big Data 5 Vs

Volume



Big Data 5 Vs

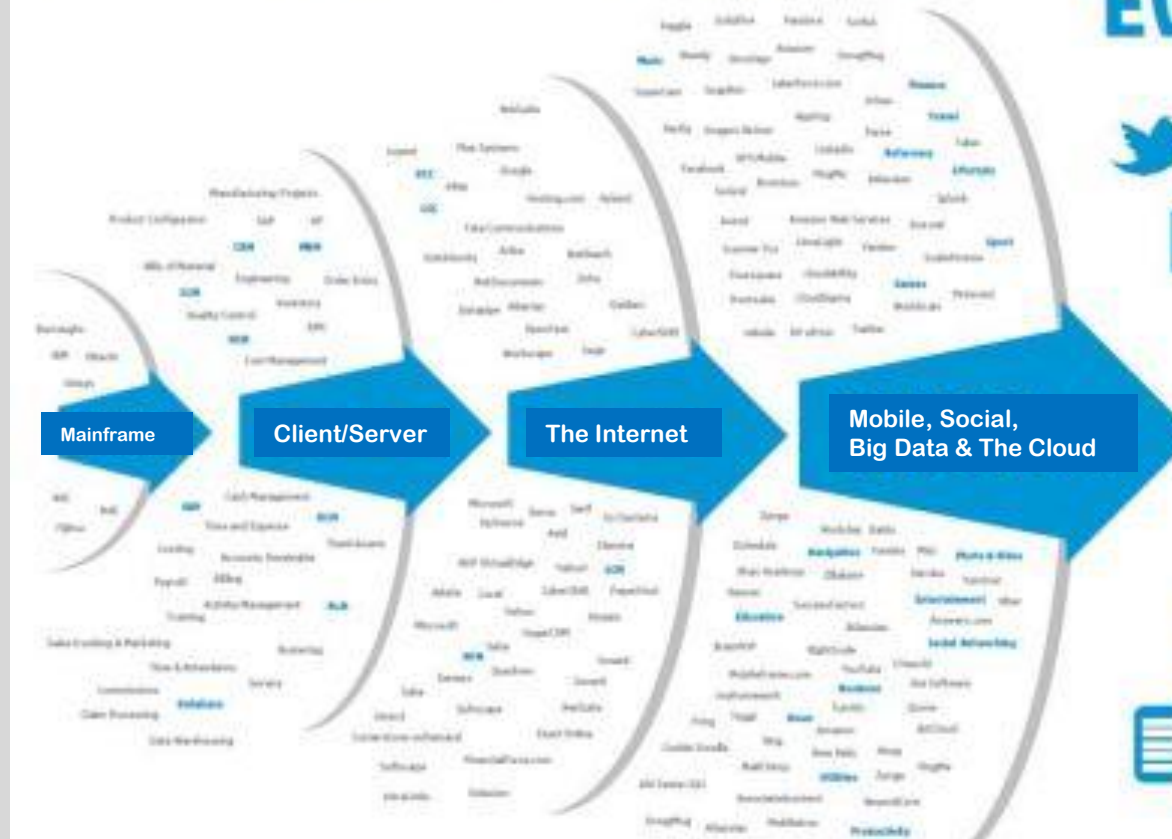
- **Variety**
 - Multiple sources
 - Multiple formats



Big Data 5 Vs

◦ Velocity

A new style of IT emerging



Every 60 seconds



98,000+ tweets



695,000 status updates



11 million instant messages



698,445 Google searches



168 million+ emails sent



1,820TB of data created



217 new mobile web users

Big Data 5 Vs

- **Veracity**
 - **Inconsistencies in the data.**

Date	Customer	Volume	Subtotal
3/08/2020	James Bond	25	USD200.00
2020. 08. 04	Potter, Harry	0.17	GBP12
Aug. 06, 2020	Emma Watson	3×10^2	AUD39.26
5/08/2020	R. Grint	??	€21,547

Big Data 5 Vs

○ Value

VOLUME

Massive amounts of data

- ✓ Social Media posts
- ✗ India's Unique Identity Database



VELOCITY

Pace, speed, immediate access

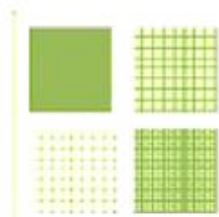
- ✓ Financial Transactions
- ✗ Census



VARIETY

Disparate sources create new insights

- ✓ Fraud Protection data
- ✗ Census



RELEVANCE

Does this insight make a difference?

- ✓ Nielsen Online Campaign Rating
- ✗ Social Media demographics (by themselves)



Emerging Big Data Ecosystem

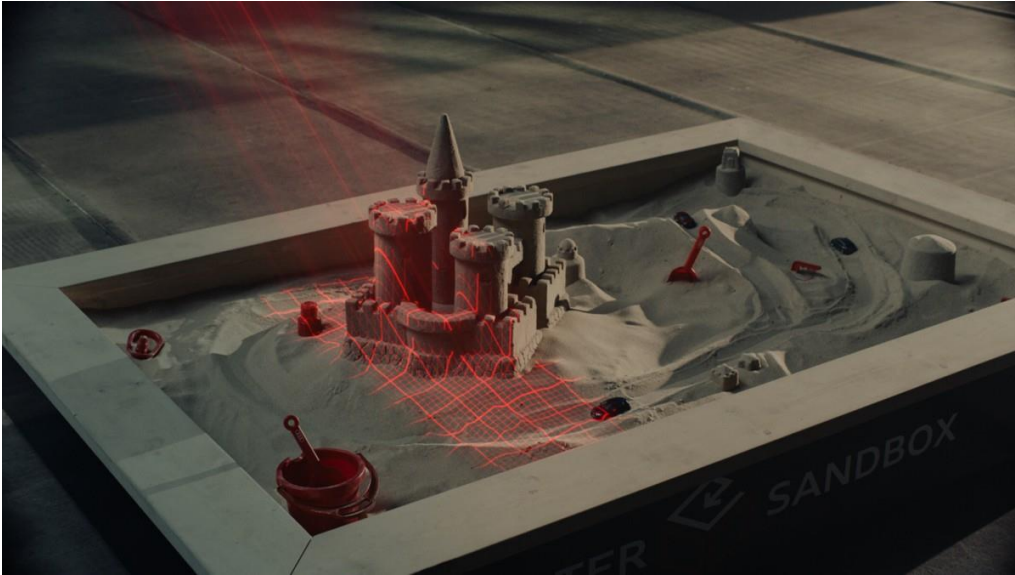
Source: Dietrich, D. ed.,
2015, Data Science & Big
Data Analytics:
Discovering, Analyzing,
Visualizing and Presenting
Data.



Big Data vs Enterprise DW/Business Intelligence

- The five Vs of Big Data will not work well with the traditional Enterprise Data Warehouse.
 - Centralised, purpose-built space. (**lack of agility**)
 - Supports Business Intelligence and reporting. (**restrict robust analyses**)
 - Analysts must depend on IT group and DBAs for data access. (**lack of control**)
 - Analyst must spend significant time to aggregate and dis-aggregate data from multiple sources. (**reduces timeliness**)
- To succeed, Big Data analytics require different approaches.





Analytic Sandbox (*Workspaces*)

- Resolve the conflict between the needs of analysts and the traditional EDW or other formally managed corporate data.
- Data assets gathered from multiple sources and technologies for analysis.
- Enables flexible, high performance analysis in nonproduction environment.
- Reduces costs and risks associated with data replication into “shadow” file systems.
- “Analyst owned” rather than “DBA owned.”

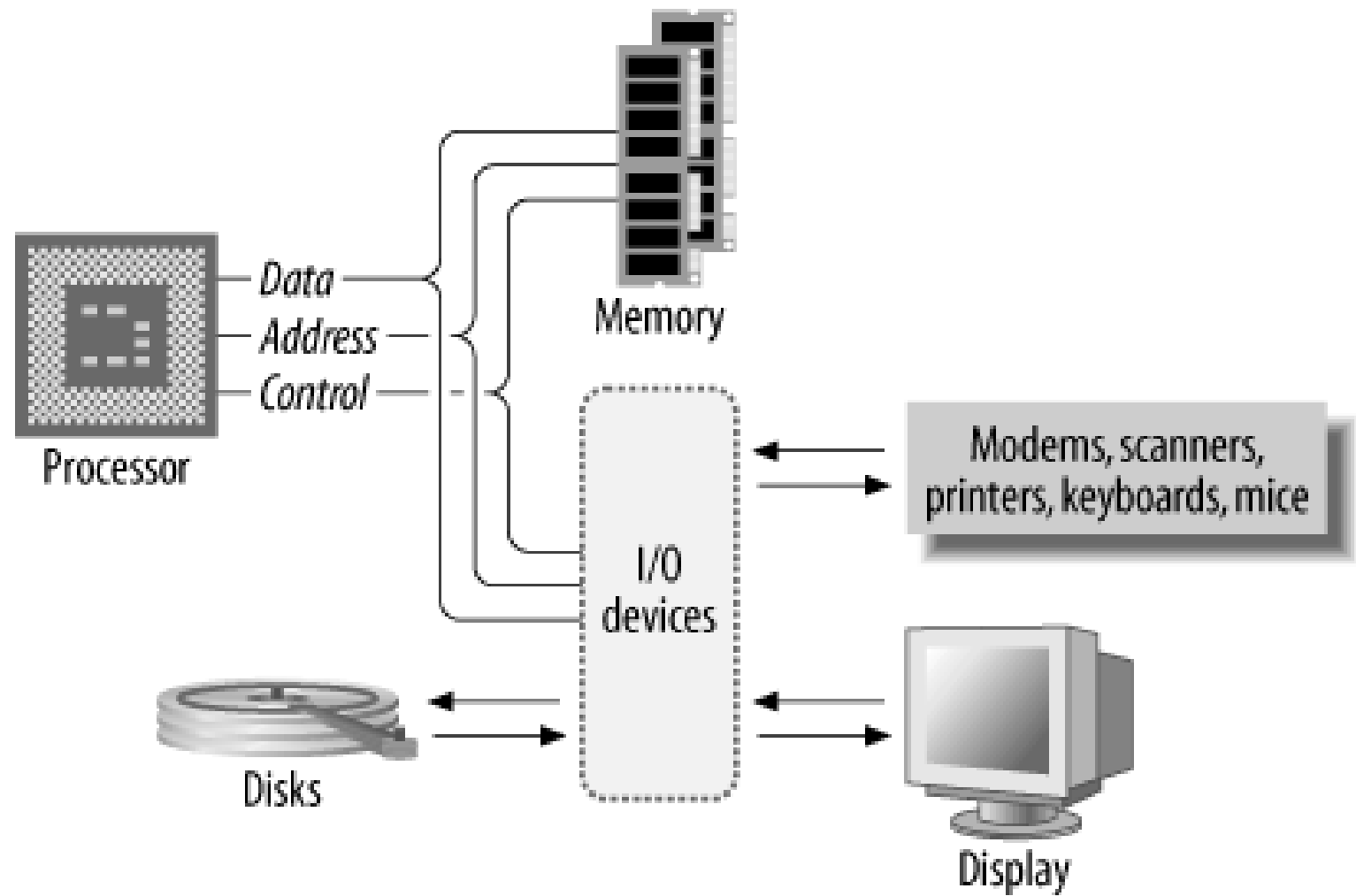


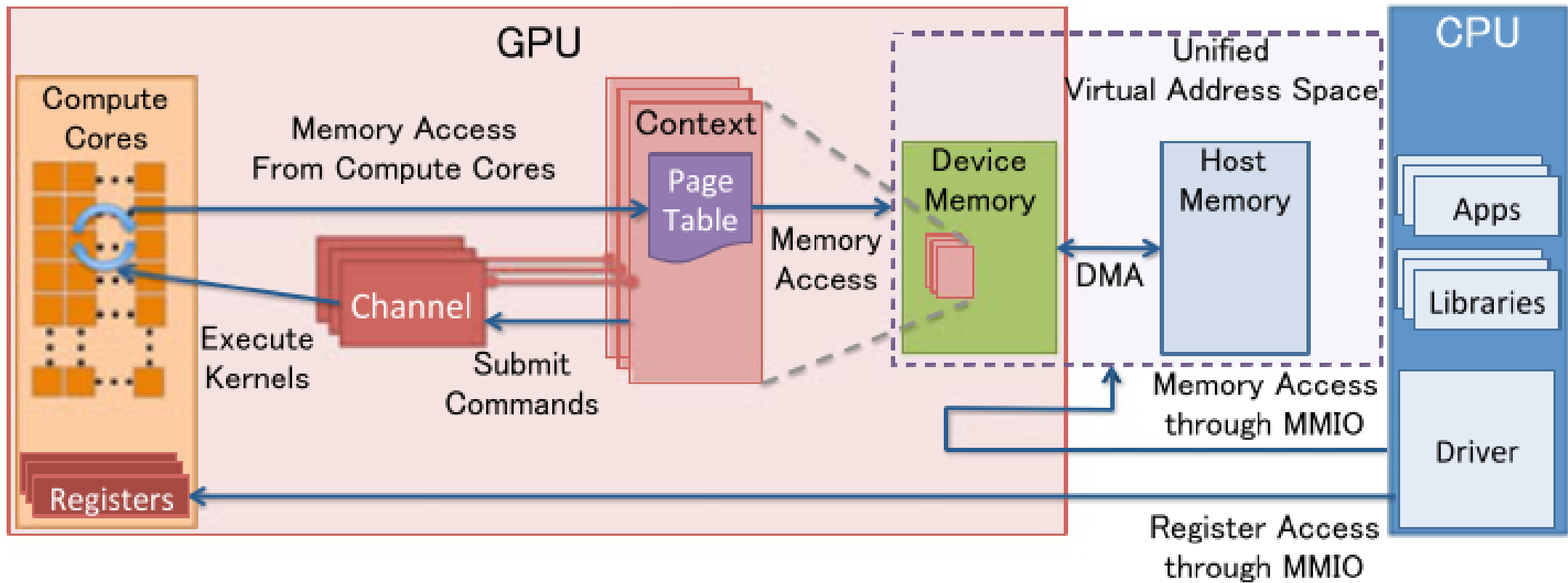
THE PLATFORM FOR STORING BIG DATA

What is Inside Your Computer?



Computer Architecture



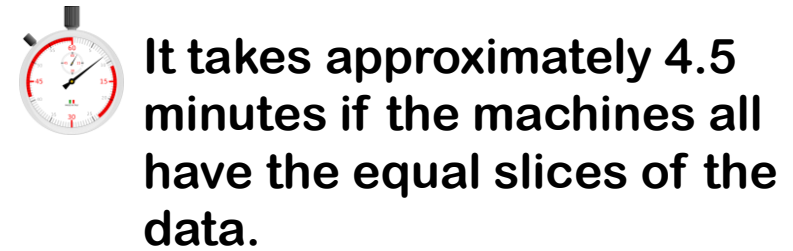


Computer Architecture: CPU and GPU Communication

- <https://insujang.github.io/2017-04-27/gpu-architecture-overview/>

A Question about Processing Time

- Assume there is a 1 GB file to be read.



Solutions for Big Data Problems



Apache Hadoop

- Apache Hadoop is a framework for running applications on large cluster built of commodity hardware.
- The Hadoop framework transparently provides applications both reliability and data motion.
- Hadoop implements a computational paradigm named Map/Reduce, where the application is divided into many small fragments of work, each of which may be executed or re-executed on any node in the cluster.
- In addition, it provides a distributed file system (HDFS) that stores data on the compute nodes, providing very high aggregate bandwidth across the cluster.
- Both MapReduce and the Hadoop Distributed File System are designed so that node failures are automatically handled by the framework.



hadoop Architecture

Application Layer



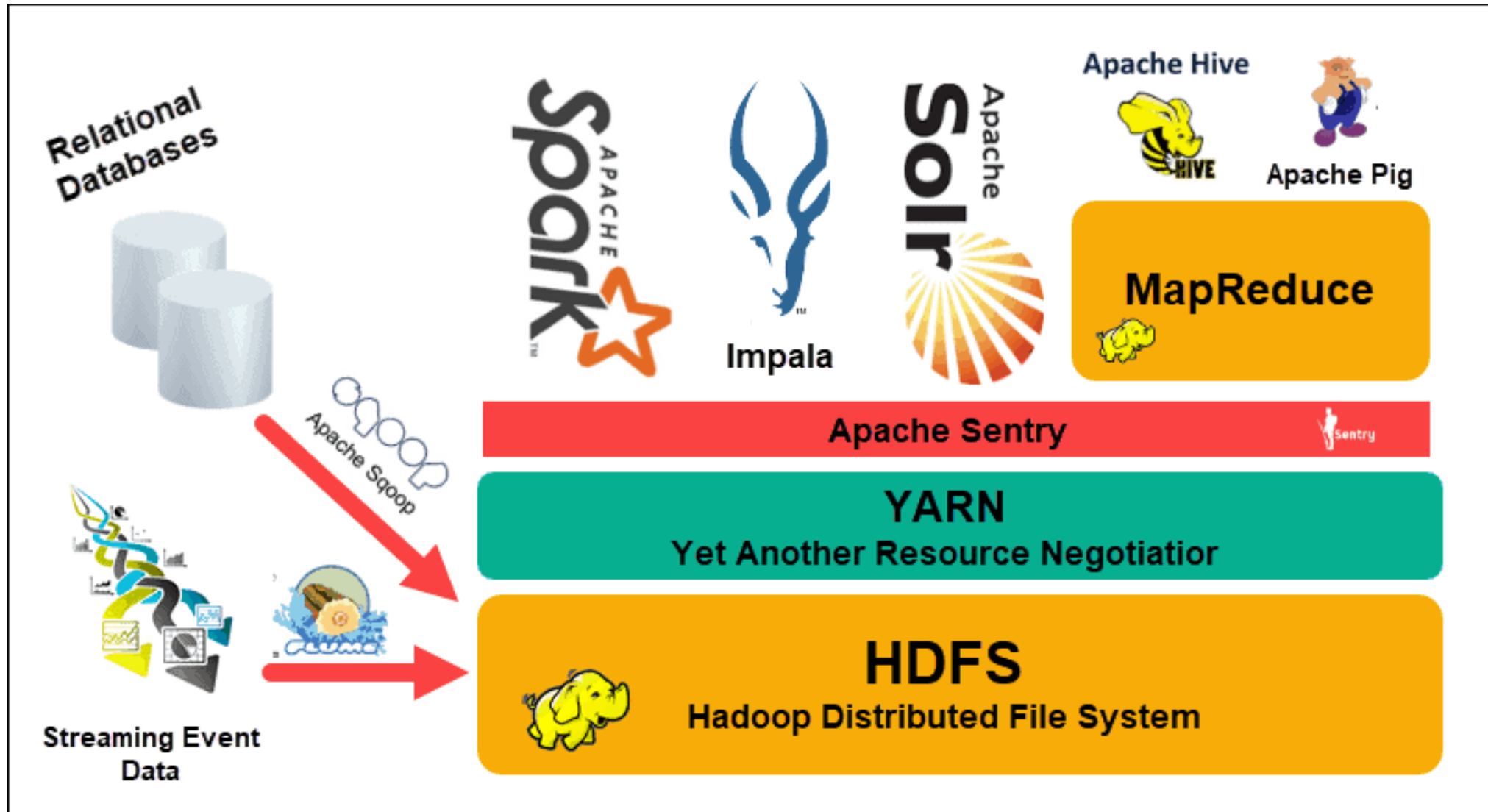
Other
Applications

Resource
Management Layer

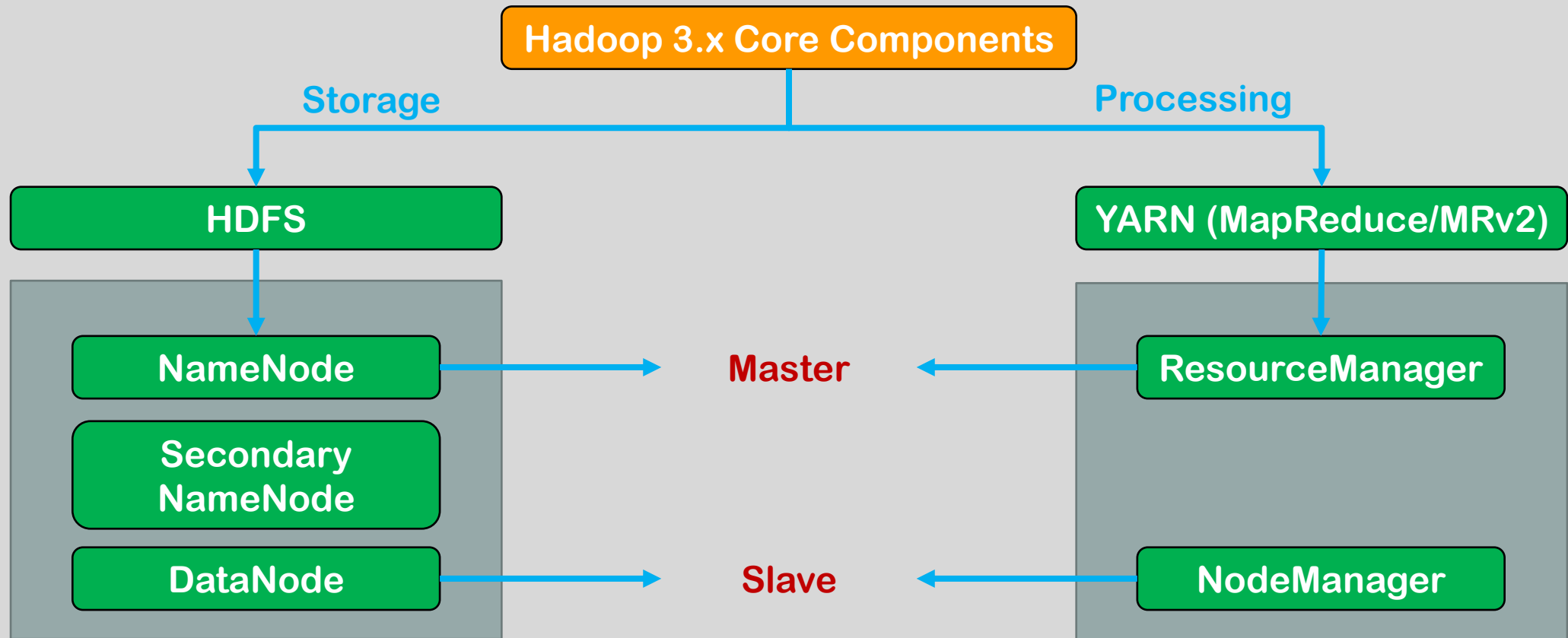


Storage Layer





Hadoop 3.x Daemons





Question

- If you have a very important file stored on your computer, what will you do to prevent the file being lost?
 - Lock it up and never access it again?
 - Make copies and store in multiple folders on a single device?
 - Make copies and store on multiple devices?

Hadoop Distributed File System (HDFS)

- The Hadoop Distributed File System (HDFS) is a distributed file system designed to run on commodity hardware.
- It has many similarities with existing distributed file systems. However, the differences from other distributed file systems are significant.
- HDFS is **highly fault-tolerant** and is designed to be deployed on low-cost hardware.
- HDFS provides **high throughput access** to application data and is suitable for applications that have large data sets.
- HDFS relaxes a few **POSIX requirements**¹ to enable streaming access to file system data.
- HDFS was originally built as infrastructure for the Apache Nutch web search engine project.
- HDFS is now an Apache Hadoop subproject.
- HDFS is designed more for **batch processing** rather than interactive use by users.
- HDFS applications need a **write-once-read-many access model** for files.

¹The Portable Operating System Interface (POSIX) is an IEEE standard that helps compatibility and portability between operating systems. Theoretically, POSIX compliant source code should be seamlessly portable.

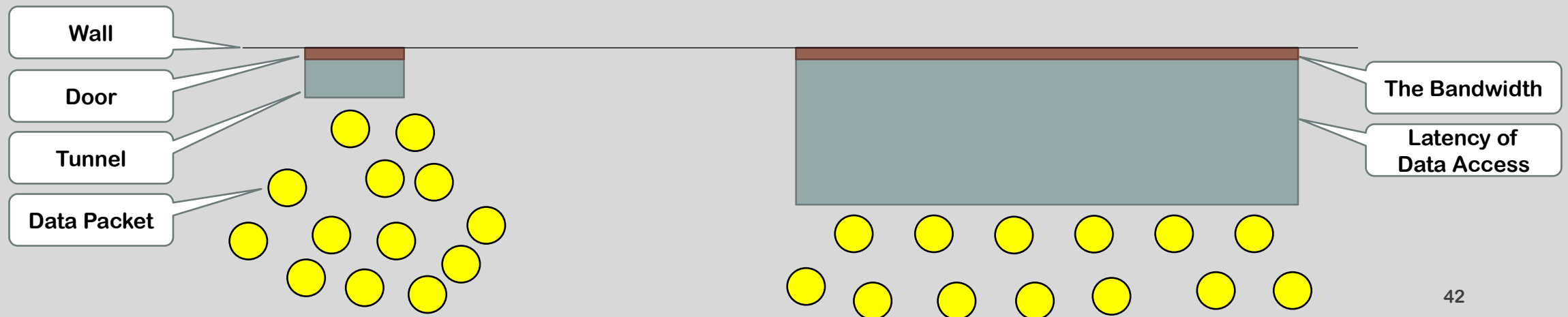
HDFS Assumptions and Goals

- **Hardware Failure**

- Hardware failure is the **norm** rather than the exception.
- **Large scale** of components with **non-trivial failure probability**.
- **Detection** of faults and quick, **automatic recovery** from them is a core architectural goal of HDFS.

- **Streaming Data Access**

- HDFS is designed more for **batch processing** rather than interactive use by users.
- The emphasis is on **high throughput** and **high latency** of data access.



HDFS Assumptions and Goals

- **Large Data Sets**

- A typical file in HDFS is gigabytes to terabytes in size. Thus, HDFS is **tuned to support large files**.
- It should provide **high aggregate data bandwidth** and scale to hundreds of nodes in a single cluster.
- It should support tens of millions of files in a single instance.

- **Simple Coherency Model**

- HDFS applications need a **write-once-read-many** access model for files. A file once created, written, and closed **need not be changed**. This assumption simplifies data coherency issues and enables high throughput data access.
- A MapReduce application or a web crawler application fits perfectly with this model.

HDFS Assumptions and Goals

- **Moving Computation is Cheaper than Moving Data**

- A computation requested by an application is much more efficient if it is executed near the data it operates on. This is especially true when the size of the data set is huge.
- This minimizes network congestion and increases the overall throughput of the system.
- The assumption is that it is often better to **migrate the computation closer to where the data is located** rather than moving the data to where the application is running.
- HDFS provides interfaces for applications to move themselves closer to where the data is located.

- **Portability Across Heterogeneous Hardware and Software Platforms**

- HDFS has been designed to be **easily portable from one platform to another**. This facilitates widespread adoption of HDFS as a platform of choice for a large set of applications.

Three Modes in HDFS

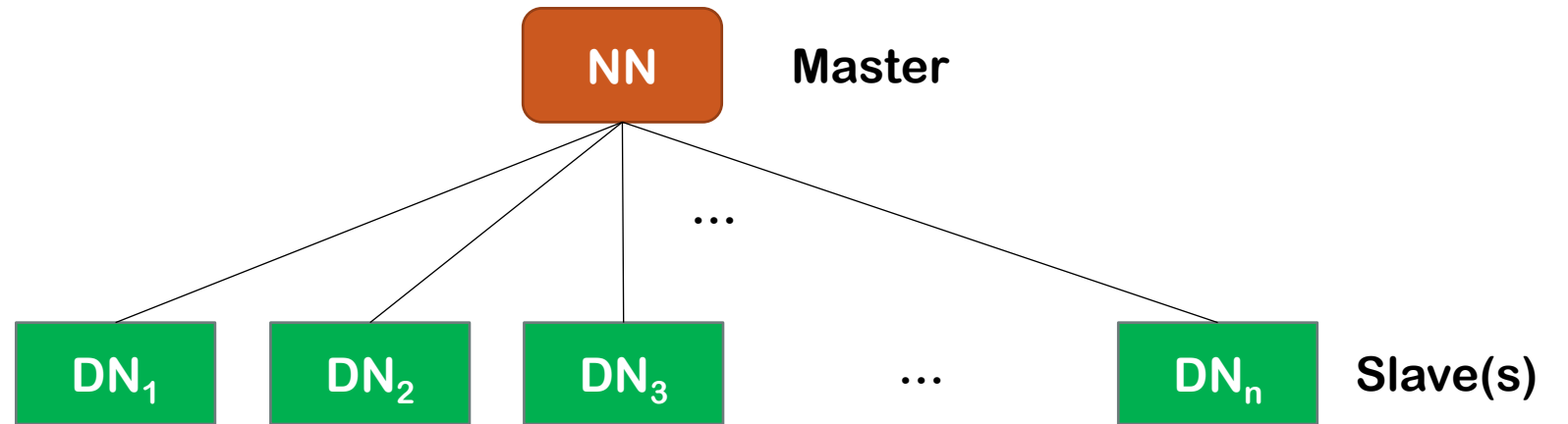
- **Stand Along Mode**
 - For testing purpose, only. **Can not be used to test real-world applications.**
- **Pseudo-distributed mode (Single-node cluster)**
 - All demands run on a single server.
- **Full distributed mode**
 - Multiple machines are used in the framework.



HDFS ARCHITECTURE

HDFS Topology

- Master/Slave architecture.
- A single Namenode (NN)
- A number of Datanodes (DN)



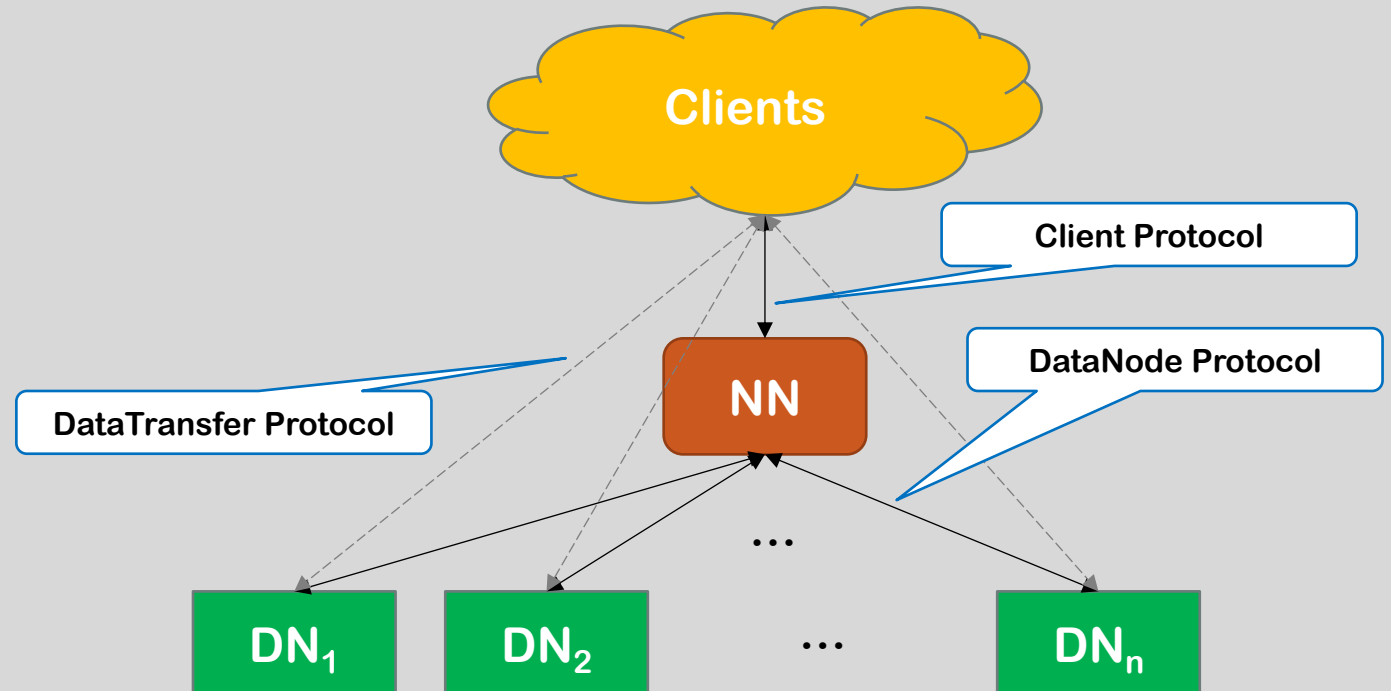
NameNode and DataNode

- **NameNode**

- Master daemon.
- Maintains and manages DataNodes.
- Records metadata, e.g. location of blocks stored, the size of the files, permissions, hierarchy, etc.
- Receives heartbeat and block report from all DataNodes

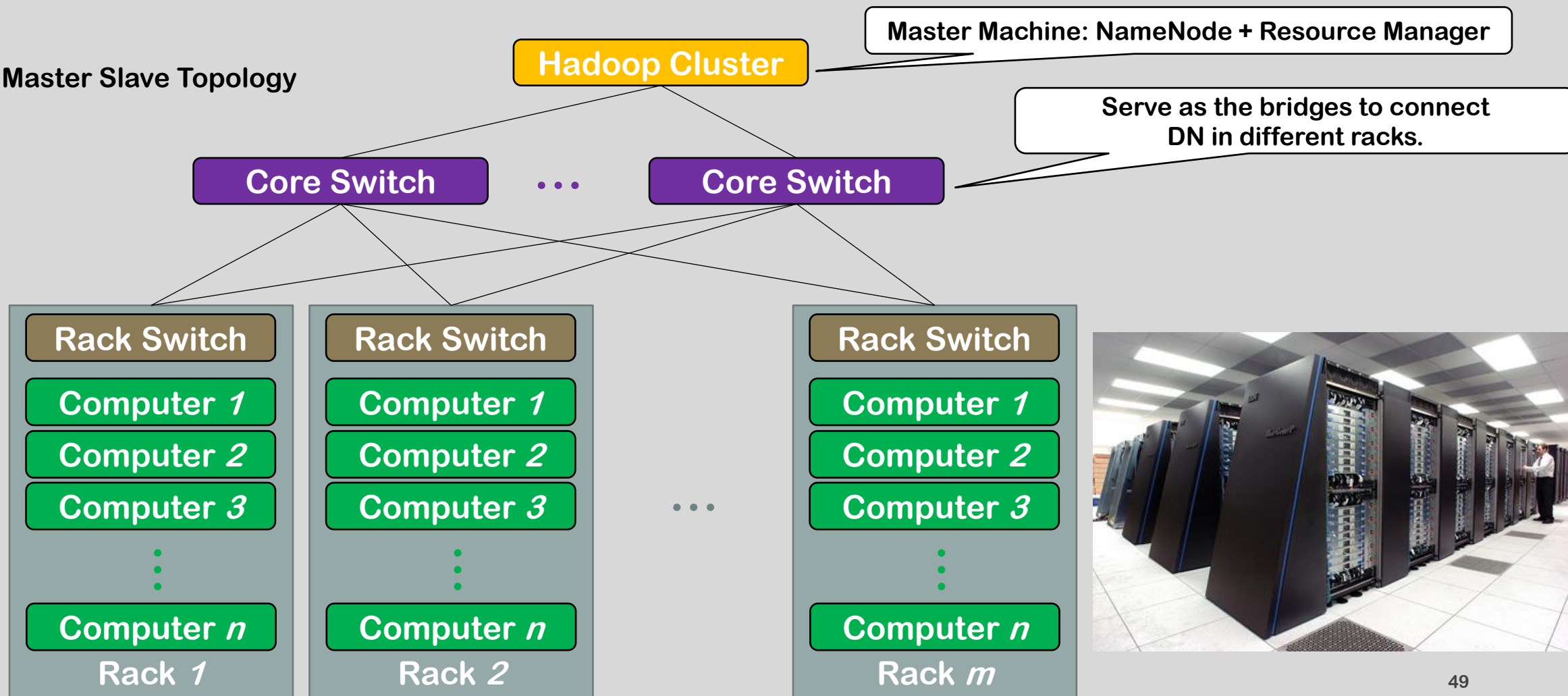
- **DataNode**

- Slave daemons.
- Stores actual data.
- Serves read and write requests from the clients.



Hadoop Cluster Architecture

- Master Slave Topology

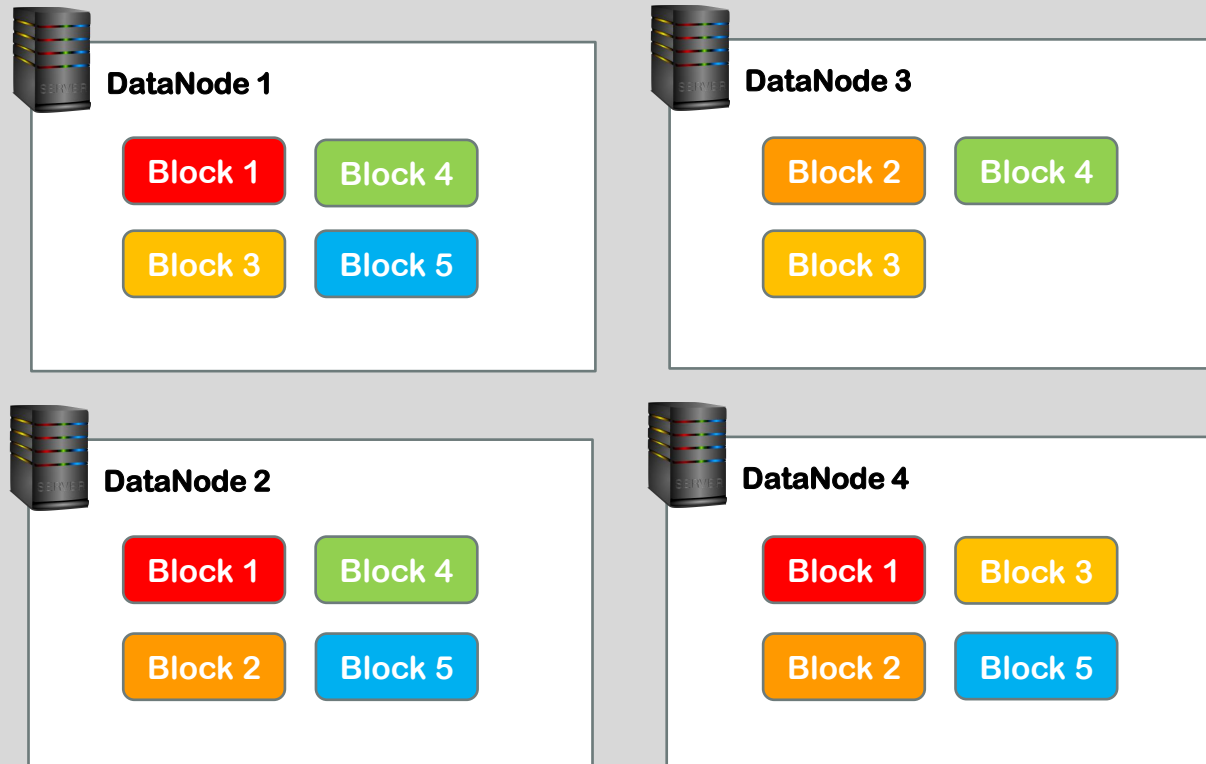


HDFS Blocks

- Each file is stored on HDFS as **blocks**.
- The default size of each block is **128 MB** in Apache Hadoop 2.x (64 MB in Apache Hadoop 1.x).
- Assume we have a file called “input.txt” of size 200 MB. It will be cut into two blocks when storing in HDFS.

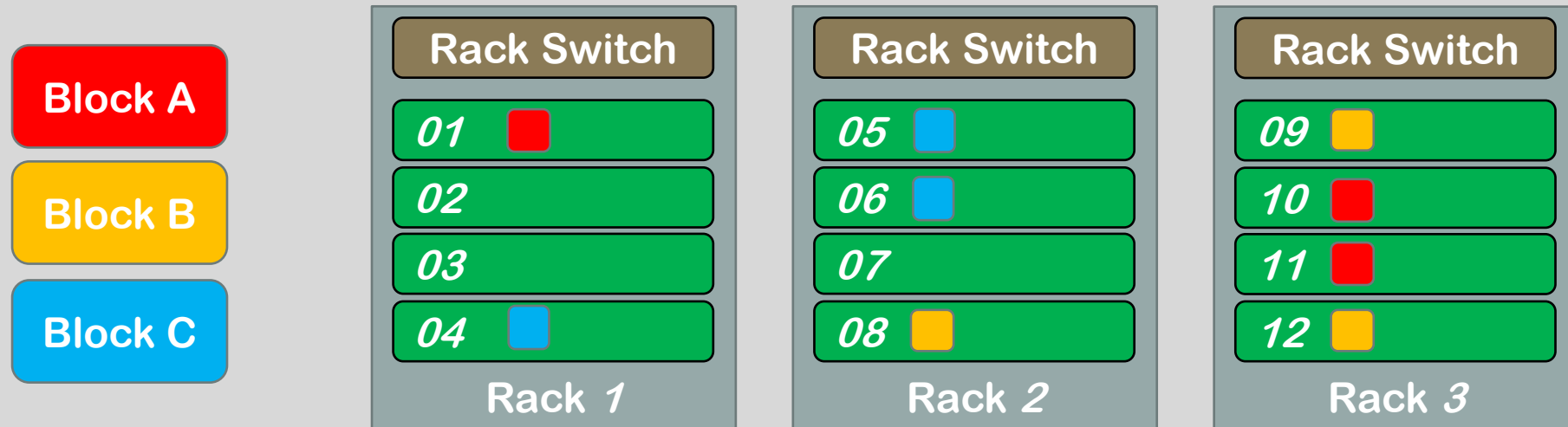


HDFS Block Replication

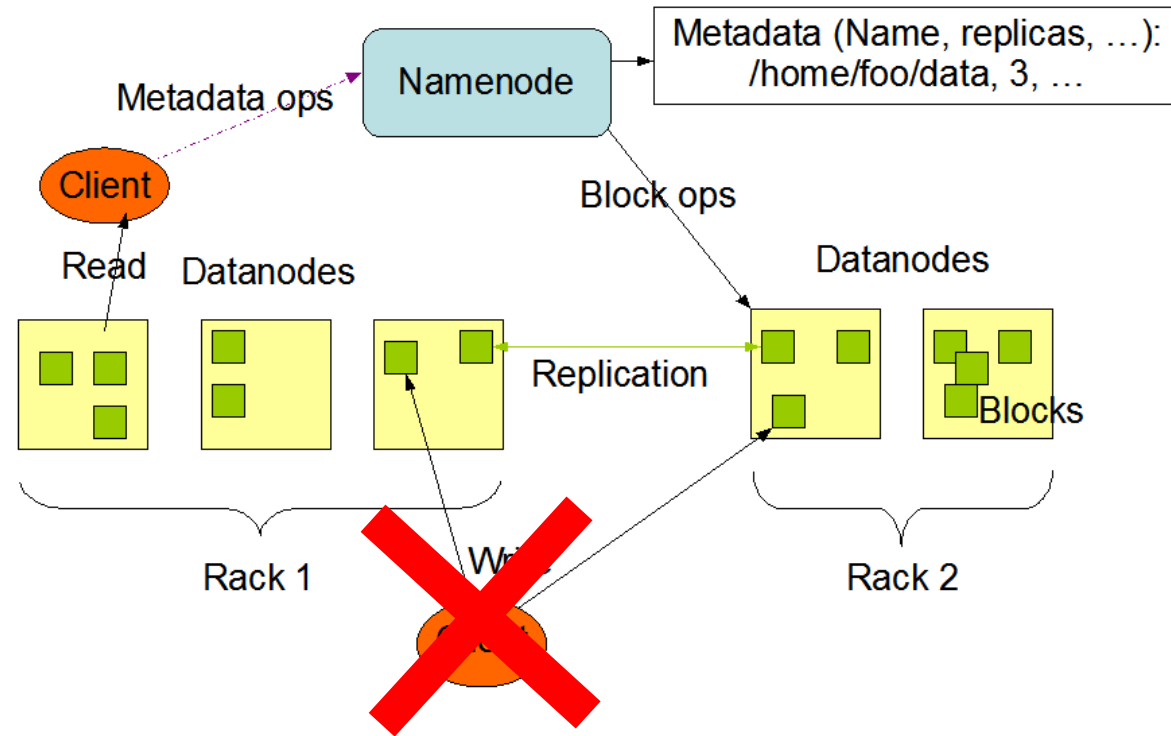


HDFS Block Replication (2)

- How does Hadoop decide where to store the replica's of the blocks?



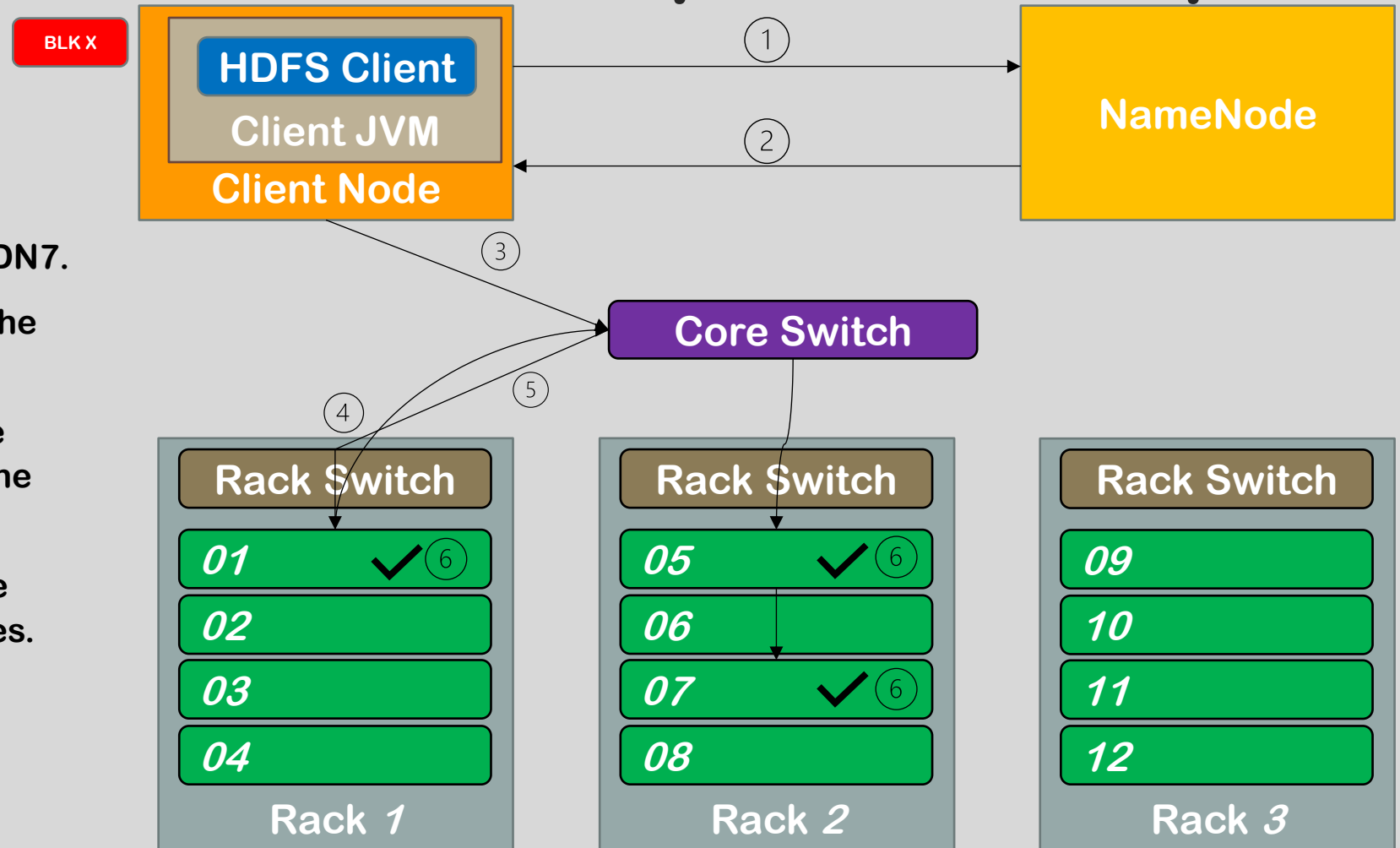
HDFS Architecture



HDFS ARCHITECTURE

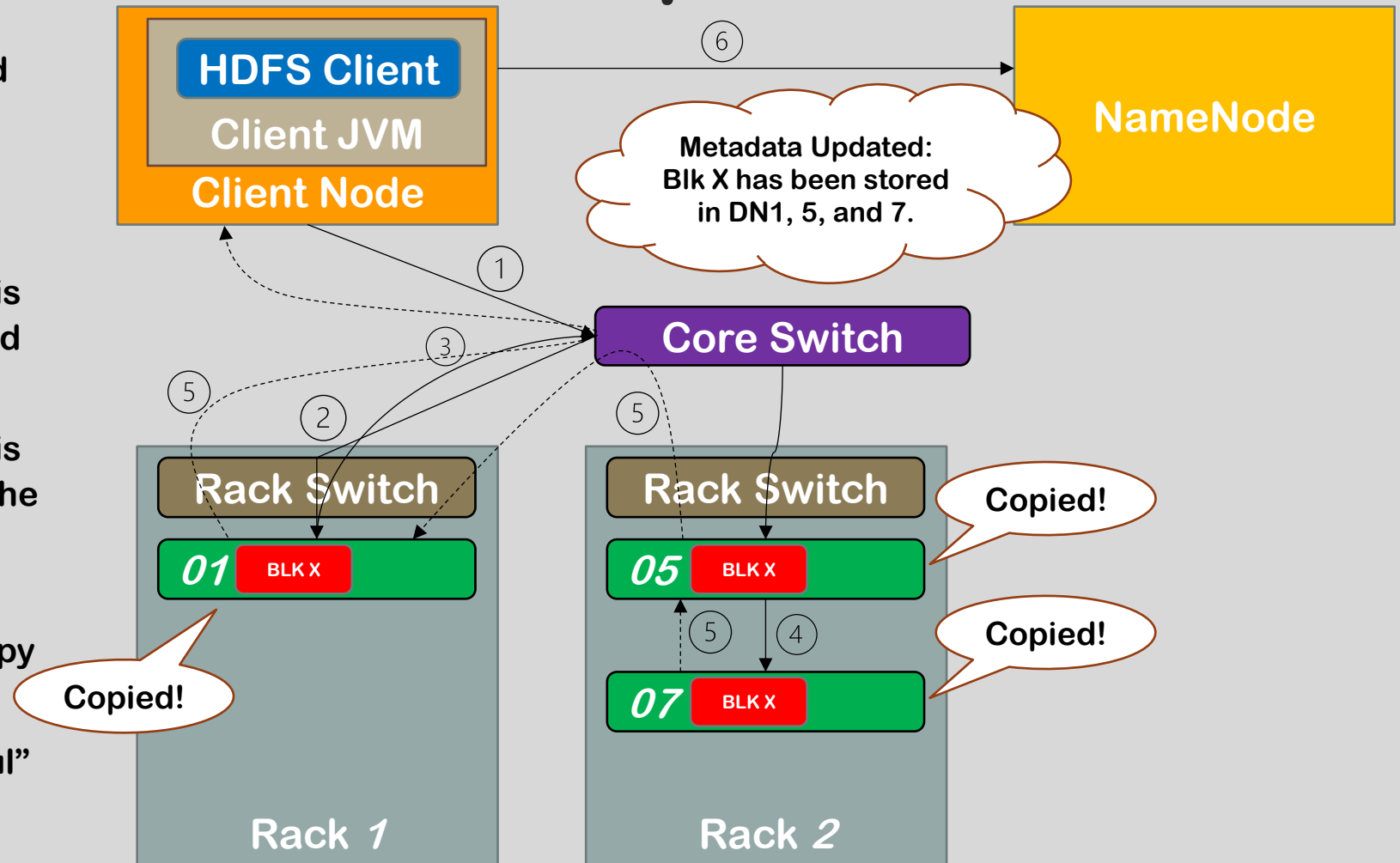
HDFS Write Mechanism – Pipeline Setup

1. Write Request – Block X.
2. IP Addresses: DN1, DN5, and DN7.
3. Sending instructions through the core switch.
4. The core switch distributes the instruction to the switches at the corresponding racks.
5. The main DataNode passes the signal to other listed DataNodes.
6. The DataNodes reply they are ready.
 - The write pipeline is created.



HDFS Write Mechanism – Pipeline

1. Write Request on DN1, DN5, and DN7.
2. Write Block X on the first DataNode.
3. Block X Replica 1 write request is sent to the second DataNode and the replica is created.
4. Block X Replica 2 write request is sent to the third DataNode and the replica is created.
5. DataNodes send back the acknowledgement of block X copy tasks are completed.
6. Client send a “writing successful” message to NameNode for updating the metadata.

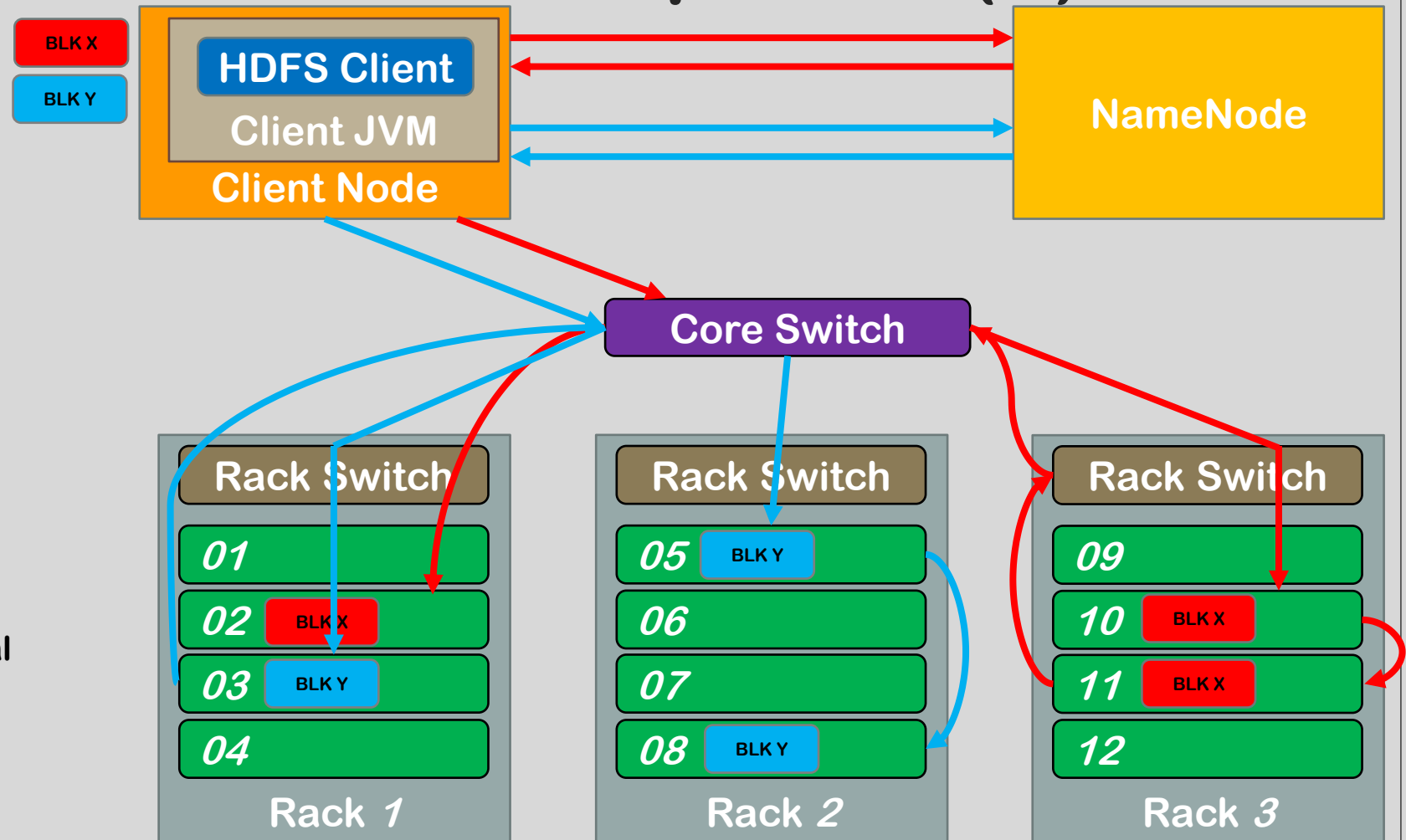


HDFS Write Mechanism – Pipeline (2)

Assume the input file is divided into two blocks (X and Y).

α GB File $\rightarrow 3\alpha$ GB Storage in HDFS and 3α GB Network Traffic.

- The pipeline itself is a sequential process.
- It can be parallel processes between pipelines.

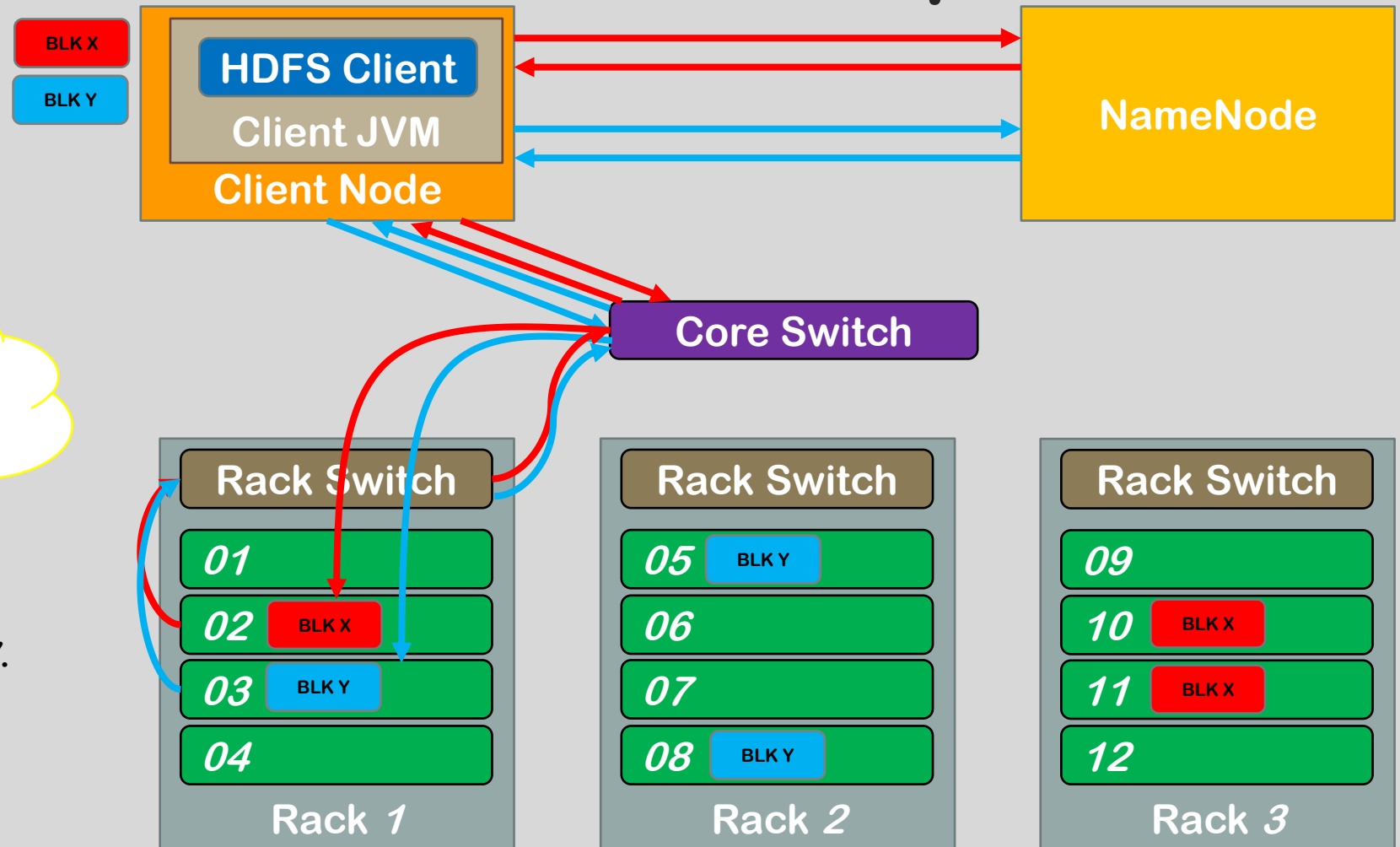


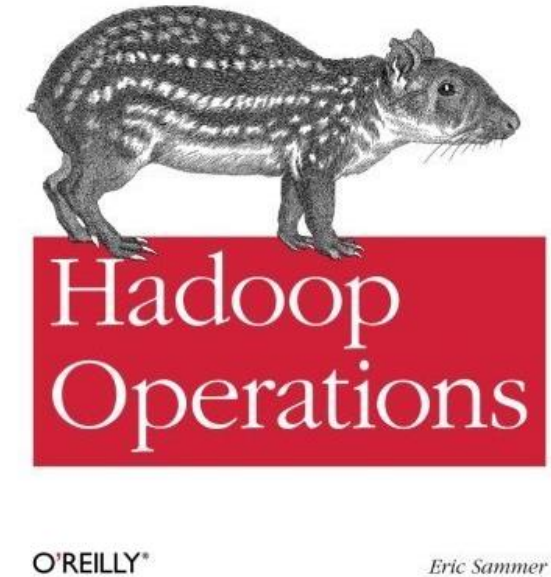
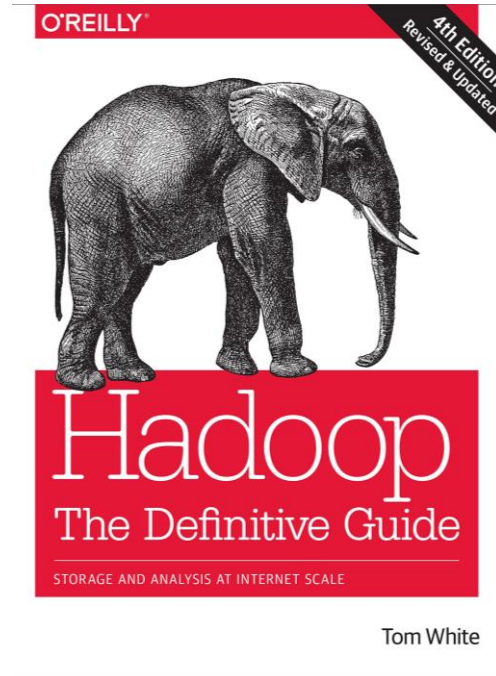
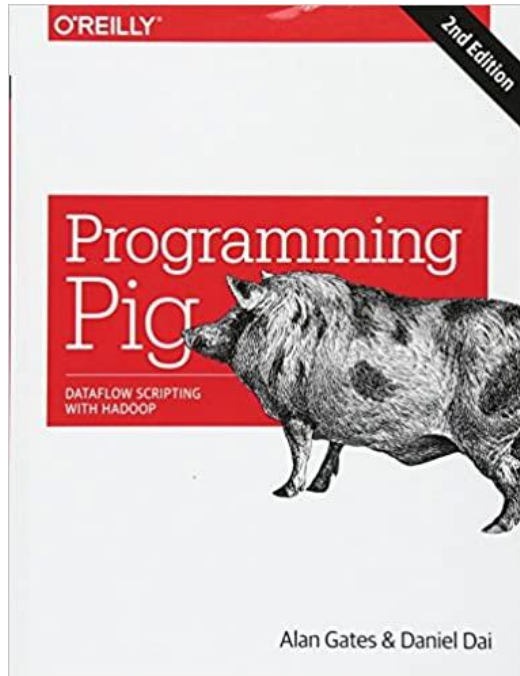
HDFS Read Mechanism – Read Pipeline

Assume the input file is divided into two blocks (X and Y).

Note: NN gives the IP addresses according to the distances between the client to the DNs, network bandwidth, etc.

1. Read Request – Blocks X and Y.
2. IP Addresses: DNs 2 and 3.
3. Read from DN2 and DN3.





Texts and Resources

- Unless stated otherwise, the materials presented in this lecture are taken from:
 - Hadoop: the definitive guide, White, Tom (Tom E.) author., 4th ed., Sebastopol, California: O'Reilly, 2015.
 - Hadoop operations, Sammer, Eric., Loukides, Michael Kosta.; Nash, Courtney.; Romano, Robert (Illustrator), illustrator., First edition., Sebastopol, CA: O'Reilly, 2012.
 - Programming pig: dataflow scripting with hadoop, Gates, Alan, author.; Dai, Daniel, 2nd edition., Beijing, China: O'Reilly, 2017



Unit Summary

- Unit structure
- Requirements, Expectations, and Rules
- HDFS

Summary

